

Visuo-Motor Remapping for 3D, 6D and Tool-Use Reach using Gain-Field Networks

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Abstract—Reaching and grasping objects in 3D is still a challenging task in robotics because they have to be done in an integrated fashion, as it is for tool-use or during imitation with a human partner. The visuo-motor networks in the human brain exploit a neural mechanism known as gain-field modulation to adapt different circuits together with respect to the task and for parsimony purpose. In this paper, we show how gain-field neural networks achieve the learning of visuo-motor cells sensitive to the 3D direction of the arm motion (3D reaching), to the 3D reaching + 3D orientation of the hand (6D reaching) and to the 3D direction of tool tip (tool-use reaching) when this new information is added to the network. Experiments on robotic simulations demonstrate the accuracy of control and the efficient remapping to the new coordinate system.

I. INTRODUCTION

Reaching and grasping objects in 3D space is still a difficult challenge for robots [1], [2] and for infants as well [3], [4], [5], [6]. It requires not only to learn how our body moves in space and where the body is (i.e., the hand), but also what the task is [7], [8] to infer its means-end [9]. Depending on the objectives, the type of grasp necessary to reach and manipulate one type of object will change; i.e., its affordance [10], [11]. For instance, whether we interact with a person or with a specific object (a coffee cup, a computer mouse or a tool), a remapping is necessary to adapt this control with respect to the object shape, its orientation and/or its relative distance to someone else [12], [9], [13].

This contextual remapping is dynamic in the brain, which permits a rapid adaptation of the body image and to new tasks as well, such as tool-use or visual imitation [14], see Fig. 1. Within the parietal and motor cortices, sensorimotor networks remap the visual transformations associated with arm motion in 3D space to align the arm to specific 3D directions toward objects [15], [16], [17], the hand to specific orientations if the task needs it [18], [19], or the arm to external spatial movement direction [20].

The underlying neural mechanism is based on gain-modulation. It confers a computational advantage, as binding across the visual, proprioceptive and motor activity is dynamic and contextual, which permits to not relearn everything from scratch for new tasks but only the necessary variations. Such approach is also advantageous toward adaptive robots in comparison with more standard machine learning methods [2] as the learning of new tasks can be continuous, probabilistic, and framed within the Bayesian framework [21], [22] for active inference [23].

Implementations of Gain-Field Neural Network (GFN) have been realized in several neurobotic groups [24], [25], [26], including ours on audiovisual integration [27], body schema [28], [29], reaching/tool-use [30], [31], [32], and handwriting [33] but not for reach with desired hand orientations and reach with tools in 3D.

In what follows, we implement GFNs to learn the visuo-motor control of a 8 degrees-of-freedom (DoF) robot arm for 3D reaching, reaching goal with desired orientation, and tool-use reaching. In experiments of simulations and of real robot, we show that this spatial remapping does not augment the complexity of the task for GFNs. At first, we present how GFNs ease the rapid learning of receptive fields sensitive to 3D locations by fusing different information. In second, we show how these multi-modal neurons construct very rapidly spatial maps to guide the arm posture to 3D locations and to 3D orientations.

The experimental section is as follows. In the first experiment, we investigate visual sensitivities of GF neurons in preferred directions, as a basis of specific firing pattern in motor cortical cells [34]. In the second experiment, in a simulation of a 8 DoF robot arm of the physical robot Reachy (Pollen Robotics), we show the performance of a first reaching neural network with GF neurons to move the robot arm in different locations in the 3D space. We describe then the properties of GF neurons to have mixed representations of preferred visual direction and certain spatial locations. In the third experiment, we test the performance of a second reaching with desired orientations of the end point GF neural network to align the hand in different 3D orientations at different 3D locations. In a forth experiment, we extended our GF neural network to tool-use for the control of a long stick at different 3D locations. Our results show that a limited number of learning samples is enough to approximate the 3D visuo-motor transformation in the working space of the robot arm with precision and with a neural architecture of minimal complexity.

II. MATERIAL AND METHODS

A. Gain-Field Mechanism

Gain-field Networks involve multiplicative interaction between neural populations from different modalities, e.g., *Vision* and *Proprioception*. By multiplying unit by unit, different afferent sensory signals can contribute proportionally to subtle adjustments of each corresponding output unit, ensuring joint distributions on resulting stimulus or impulse (priors and cues). The paradigm in Fig. 1(b) (I) elucidates that a 3D reach can be predicted by multiplying

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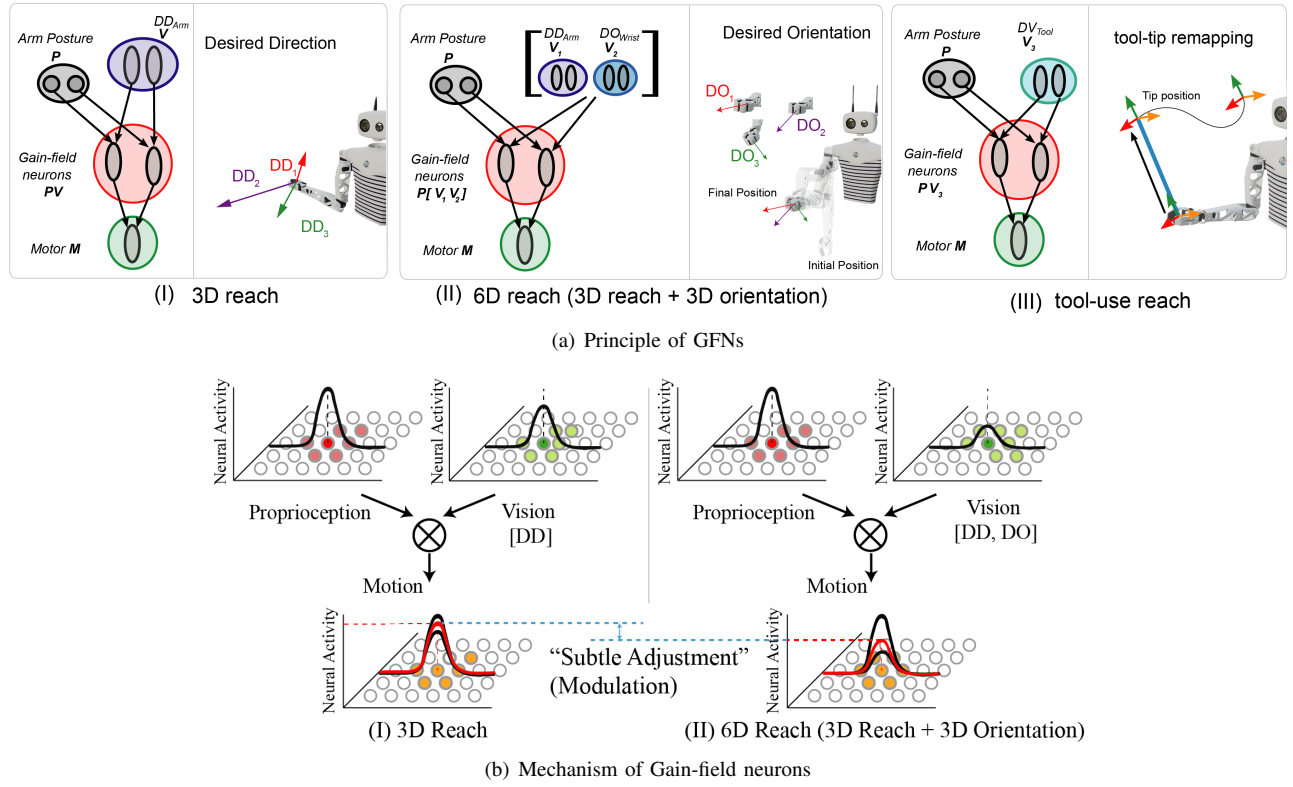


Fig. 1. (a) GF neurons found in the brain are neurons that bind incoming signals (e.g., visual, motor and proprioceptive) through multiplication (see (b)). At the population-level, they can guide the reaching movement of the arm toward desired direction (I), control the wrist in a particular orientation (II) during reaching, or integrate external tool-tip visual information into internal models (III). (b) GF neuronal responses are modulated dynamically and collectively by different sensory signals. Similar to a Bayesian process, GF neurons adjust conditionally their output based on prior information. Such a resulting subtle adjustment describes how a GFN is sensible to *supplemental* information related to various tasks.

the posture and displacement vector of the robot arm. When integrating desired gripper orientation into visual information as depicted in Fig. 1(b) (II), the robot is able to adapt the orientation as approaching a target, revealing that the desired orientation is taken into account and thus perturbs firing rates of these motor sensitive cells [35].

Mathematically speaking, such sensorimotor signals V (*Vision*) and P (*Proprioception*) can be regarded as probability distributions. Therefore, the multiplicative relationship between input signals and the output M (*Motion*) can be described as:

$$X_j = \sum_{i,j=1}^{I,J} W_{ij} \cdot P_i \quad (1)$$

$$Y_j = \sum_{k,j=1}^{K,J} W_{kj} \cdot V_k \quad (2)$$

$$M_n = \sum_{n,j=1}^{N,J} W_{nj} \cdot (X_j \cdot Y_j) \quad (3)$$

B. Networks Architecture

To illustrate that the spatial remapping does not augment the complexity of GFN, we applied the same structure for all

3 GFNs: 6 layers in total without counting the input layer, composed of 1 fully connected layer after each input layer (both have 16 nodes for equalizing the input dimension thus $J = 16$ in Eq. (1)-(3)), 1 element-wise multiplication layer, 3 hidden layers (with 32 hidden units fully connected with each other) and 1 output layer. Training results of different numbers of hidden layers and units will be further analysed in III.

C. Reachy Robot

The Reachy robot 2019 (Pollen Robotics) was used in our experiments, both for simulations and for tests. Reachy is an open-source humanoid robot with the 8 DoF human-like robotic arm and can reach 827.5mm furthest horizontally [36]. With Robotis Dynamixel servomotors and embedded sensors, Reachy can provide high-level motor information such as joints angular positions, thus facilitating the creation of training data.

D. Inputs and Output

Different from most artificial neural networks, GFNs have more than one input. One of the two inputs P , as an 8-dimensional vector corresponding to 8 DoF of Reachy, is the robot's initial posture/joint angles. Notice that, the location and the orientation of end-effector can be calculated by the joint angles of robot arm based on Forward Kinematics (FK) [37].

What distinguishes Gain-field networks from each other is the second input V , as detailed in Table I. In comparison to 3D reach, an additional visual component “orientation” is concatenated into visual input as a conditioning signal for 6D reach, as depicted in Fig. 1 and Fig. 1(b). What’s more, in the tool-use reach, the visual input becomes the external displacement vector of tool-tip instead of the internal displacement vector of end-effector as in 3D reach.

The output motion M is an 8-dimensional vector composed of the final joint angles of robot arm. How the inputs and the output are implemented to motion loop algorithm applied in section III will be outlined in the next subsection.

E. Motion Loop Algorithm

Specifically, in our experiments, the robot is instructed to execute one-tenth of joint angle displacement in each iteration. Therefore, the relation of output *Motion* between two successive iterations can be described by:

$$\begin{aligned} P_{t+1} &= (M_t - P_t) \times 0.1 + P_t \\ &= (V_t \otimes P_t - P_t) \times 0.1 + P_t \end{aligned} \quad (4)$$

where $\otimes$ represents the prediction of GFNs. In practice, given target information (position or position+orientation) and by obtaining initial joint angles from robot directly, the initial position of end-effector can be calculated (by Forward Kinematics) and so does the displacement vector. Setting the repeat condition that the Euclidean distance between end-effector and target keeps getting shorter (or equalized within 5 iterations) overtime, this algorithm will generate an array of joint degrees that can be executed by robot.

F. Training Data

As stated in section II-C, Reachy records synchronously joint angles of which we can take advantage for data sampling. Proprioceptive input and motion output are both raw sampled data joint angles. Differently, visual inputs require FK calculation as mentioned in II-E.

1) *3D Reach GFN Training Data*: We sampled the training data by recording postures as manually moving Reachy’s end-effector in an array of directions resembling a “3D star pattern” (or a 3D Cartesian coordinate system). As moving robot arm manually, we can have a real-time feedback as soon as the robot arm reaches its limit. Therefore, the path length of different directions varies. What differs between “star” groups are central starting positions (just like the origin point in the 3D Cartesian coordinate system). Therefore, In total, 6 “stars” have been recorded and 13134 data were collected. Notice that, during the manipulation, the gripper’s joint angle remained constant as it is irrelevant for the end-effector position. FK calculations of the visual info was done afterwards. We generated more training data by reversing the displacement vector and exchanging the input and the output accordingly in order to improve the capacity of the GFN models. As a result, totally 26268 data were used for the model training.

2) *6D Reach GFN Training Data*: Different from the previous model, where the gripper’s orientation was not taken into account, in this case, postures were recorded by manually moving end-effector in a “star” as in section II-F.1 but each time the gripper orientation was rotated approximately (not strictly due to human error and it is acceptable) to the orientations listed in the footnote of Table. I. Therefore, 6 groups of data were collected, corresponding to 6 desired orientations. We sampled 33581 data in total and a similar process has been done to generate more data as in section II-F.1. Note that, besides reversing the displacement vector of visual input, desired orientation should be replaced by the initial orientation as well. In sum, 67162 data were used for training.

3) *Tool-use Reach GFN Training Data*: We took advantage of the same training data of the reaching model and translated only the direction vector of the robot’s end-effector into the direction vector of the tool (tool_vector). Therefore, the size of the dataset for the tool-use reaching model is the same as the 3D reaching model. For our purpose, the robot holds a tool with the length of *length_tool* and the tool is perpendicular to the end-effector in the initial state, we can then regard it as a translation transformation and use a homogeneous transformation matrix to determine the position of a tool tip given. As the tool is perpendicular to the gripper, we can consider it as the gripper moves along the axis X, and the distance of this translation equals the length of the tool. Hence the 4x4 matrix below:

$${}^{tool}_{gripper}T = \begin{bmatrix} 1 & 0 & 0 & tool_length \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (5)$$

From it, we can obtain the coordinates of the tool tip in the end-effector-centred reference frame. To represent it in the original reference frame of the robot, what we need is just a simple inner product of two matrices:

$${}^{original}_{gripper}T = {}^{original}_{gripper}T \cdot {}^{tool}_{gripper}T \quad (6)$$

where ${}^{gripper}_{ori}T$ is also a 4×4 matrix that can be obtained based on Forward Kinematics calculation. With these data, we can then process model training.

G. Network Training

The learning stage was done using Eq. 3. Learning in the networks was accomplished by using the stochastic gradient descent based on supervised learning and setting learning rates to 10^{-3} . RMSprop was applied to optimize the training loss with rectified linear unit (ReLU) as activation functions and mean squared error (MSE) as the loss function. Setting the batch size at 128 and with 1000 epochs, neural networks managed to converge. An additional step for the 6D reaching model is to normalise the input V because the ranges of the

TABLE I
VISUAL INPUTS V IN DIFFERENT TASKS.

Task	Visual Input V	Mathematical Representation
3D Reach	{Direction Vector}	$\{X^* - X_t^{EE}, Y^* - Y_t^{EE}, Z^* - Z_t^{EE}\}^2$
6D Reach	{{Direction Vector}, {Desire Orientation}}	$\{X^* - X_t^{EE}, Y^* - Y_t^{EE}, Z^* - Z_t^{EE}\}, \{Ort_x^*, Ort_y^*, Ort_z^*\}^3$
Tool-use	{Direction Vector of Tool}	$\{X^* - X_t^{Tool}, Y^* - Y_t^{Tool}, Z^* - Z_t^{Tool}\}^4$

² X^*, Y^*, Z^* denotes the Cartesian coordinates of the defined target. $X_t^{EE}, Y_t^{EE}, Z_t^{EE}$ denotes the Cartesian coordinates of the end-effector at instant t .

³ Ort^* denotes the 'xyz' representation of Euler angles of desired orientation in degree. Details of desired orientations for the 6D reach experiment in III-C are: $[0^\circ, -90^\circ, 0^\circ]$, $[0^\circ, -45^\circ, 0^\circ]$, $[0^\circ, -135^\circ, 0^\circ]$, $[-90^\circ, 0^\circ, 90^\circ]$, $[-45^\circ, 0^\circ, 90^\circ]$, $[-135^\circ, 0^\circ, 90^\circ]$.

⁴ $X_t^{Tool}, Y_t^{Tool}, Z_t^{Tool}$ denotes Cartesian coordinates of tool tip at instant t .

two features are very different: desired orientations vary from $[-180, 90]$, displacement vectors vary from $[-0.65, 0.71]$.

III. EXPERIMENTS

Adopting the gain-field mechanism presented previously, we can then exploit their computational capabilities to conduct 3D reach, 6D reach, and tool-use reach based on sensory modalities. Before analysis of GFNs' competencies on these tasks, it is essential to analyse if GFNs have learned well at the initial stage. Therefore, we characterize the overall performance of networks by comparing errors before and after training as shown in Table II. Notice that the initial Euclidean error represents the distance between target and the initial end-effector position (before training), whereas the updated Euclidean error represents the distance between target and the updated end-effector position (after training). Orientation error is calculated with Eq. 7. Both the updated position and the updated orientation can be calculated by Forward Kinematics of joint angles.

$$\sum_{i=1}^{I=3} = |Ort_i^{initial} - Ort_i^{updated}| \mod 180 \quad (7)$$

As can be seen, the mean and median of euclidean errors before training locate around 0.2, whereas after training, the means and the median is decreased by 10 times its value. Similar to orientation errors, the mean and median of orientation error have dropped considerably after training, from approximately 172 degrees and 191 degrees to 25 degrees and 6 degrees, as indicated in Table. II. These results confirm the good average performance of the network. Furthermore,

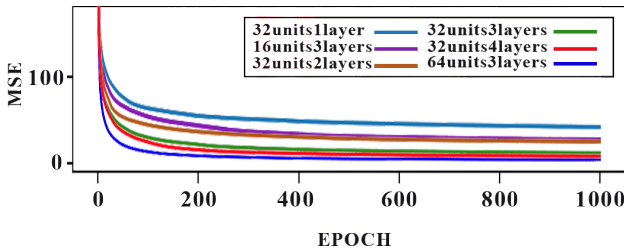


Fig. 2. Error curves during training: x-axis indicates number of epoch trained; y-axis indicates the training error mean squared error (MSE)

we initially trained 6 networks of 6D reach model with the following increments of hidden unit number: 16, 32, 64 and of hidden layer number: 1, 2, 3 and 4. Fig. 2 shows the error

curves generated by a representative network of each size as indicated by the associated label. The y-axis represents the mean squared error (MSE) during training, whereas the x-axis indicates the number of epochs. Networks with a greater number of hidden units and/or hidden layers did show better loss results. However, the performance of GFN depends on the training data as well. Therefore, we focused on networks with 32 hidden units and 3 hidden layers for the following experiments as they are the minimum numbers to have the maximum MSE deviation of two neighboring combinations of hyper-parameters.

Training loss of 3D reach and tool-use reach model with same hyper-parameters are indicated in Table. 2. In the rest of this section, we will analyse the results of a series of designed experiments to determine the GFNs' effectiveness and efficiency in manipulations of the robot arm specified by task requirements.

A. Gain-Field Neurons Visuo-Motor Learning

To begin with, an experiment designed to explore an important property of GF neurons referred to as preferred direction (PD). For brevity, only the reach model was included in this experiment. The first step was to generate another 64 dataset, namely 128 (64 as initial posture and another 64 as final posture) 8-dimensional joint angle vectors through online simulation in order to have a random distribution of 64 initial positions and 64 targets in Reach's workspace. These data were then converted into inputs as described in section II-D, and fed into the trained reaching GFN. As stated above, the applied architecture has 3 hidden layers and each has 32 hidden neurons. Therefore, to elucidate the preferred direction property of GF neurons during reaching, we focused on the neurons in the last hidden layer and plotted directions in which neurons showed the highest activities.

Result of the preferred directions of the 7th neuron is shown in different viewing angles in Fig. 3. The color saturation of arrows is positively correlated with neuron activity. It is notable that at some positions, no arrow can be found, which is because when departing from these prescribed positions, the activity of the neuron approximates zero. Based on the result, we claim that the 7th neuron has a strong preference for guiding the robot arm towards X- direction when starting from these positions. Notice that, the preferred direction of neurons depends also on the training data, which means that the 7th neuron may show no/other preference

TABLE II
EUCLIDEAN ERRORS AND ORIENTATION ERRORS BEFORE AND AFTER TRAINING

Task	Initial Euclidean Error			Updated Euclidean Error			Loss (MSE)
	Range	Mean	Median	Range	Mean	Median	
3D Reach	[0.0, 0.549788]	0.186952	0.187473	[0.000332, 0.059689]	0.015276	0.014246	8.7752
6D Reach	[0.004305, 0.839520]	0.306247	0.294505	[0.000412, 0.279702]	0.024997	0.021752	11.1598
Tool-use Reach	[0.0, 0.55145]	0.191712	0.192510	[0.000772, 0.068552]	0.016808	0.015736	10.0715
6D Reach	Initial Orientation Error			Updated Orientation Error			
	Range	Mean	Median	Range	Mean	Median	
	[0.28, 413.98]	172.86	191.75	[0.06, 374.77]	25.83	5.99	

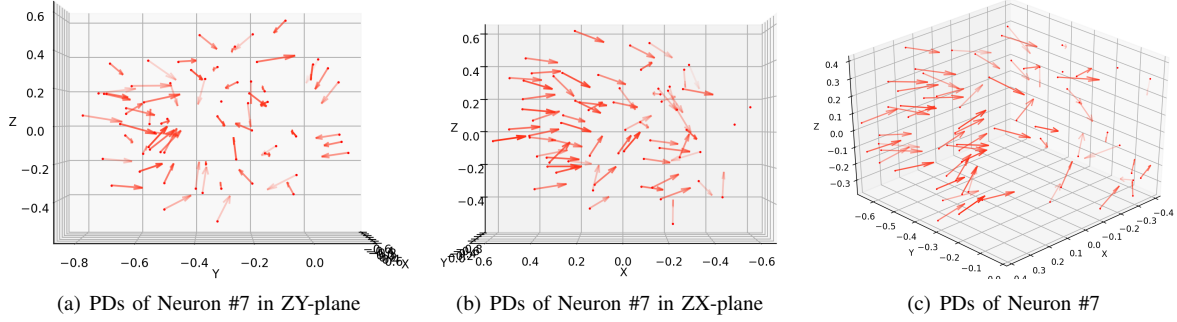


Fig. 3. The 7th neuron in the last hidden layer of 3D reach model shows preference in guiding the robot arm toward X- direction.

in directions if the model is trained with other data. Such a property implies their roles involved in the visuo-motor control, as interpreting constantly and continuously visual and proprioceptive signals into specific firing patterns to generate specific motor schema. This property will be further discussed in the next section.

B. Visuo-motor Transformation for 3D Reaching

This experiment is designed to identify the capacity of GF neurons in providing a sequence of specific predictions of 3D reach. The robot was instructed to reach target points selected from an array of directions resembling a “star pattern” that were equally 0.2m distant from the central initial position. Information about the target was passed into the neural network, allowing the execution phase loop activity. The proprioceptive input serves to inhibit loop activity and eventually stops arm movement as outlined in section II-E.

Results are shown in Fig. 4, where 3 experimental initial positions are [0.4711, -0.1773, -0.1744], [0.3942, -0.3377, -0.0194], [0.3951, -0.3974, 0.0706]. As can be observed, robot’s performance varies in different directions. Taking the blue star in Fig. 4(a) as an example, different from attaining an object in X+ direction, which has an error of 0.0664m at the 44th iteration, the robot can approach better in Y-direction, with an error of 0.0177m at the 53rd iteration. The discrepancy in performance is due to the fact that the position with better performance is more learned than another one and thus has a better ability to generalize. Nevertheless, the robot arm is able to depart from the correct direction and finally reach the target with 1.5cm precision on average within 80 iterations.

In Fig. 4(b), targets in blue, red, green, purple, black and brown are placed in Z+, Z-, Y+, Y-, X+ and X- directions

relative to starting positions respectively. As a supplement to the PDs of the 7th neuron discussed previously, we plotted neural activity patterns in Fig. 4(c) throughout the black path (left) and the brown path (right). As demonstrated, when moving towards X+ direction, the 7th neuron remained inactive whereas towards X- direction, the neuron had higher activity at first and converged to zero as the location changed.

Results of these simulations indicate that the multiplication neural model has the competency to guide the robot arm towards the desired direction and finally reach the object. Based on these results, in the next section, we increase the difficulty of tasks by taking orientations into account to probe how GF neurons react to this additional visual signal.

C. Gain-modulation of Supplemental Information for 6D Reach

The previous experiments were resolved with the estimation of the motion information by multiplying direction as *vision* and posture as *proprioception*. Differently, in this experiment, desired orientations will supplement to tune the resultant motion response and generate more complex movement configurations. The task consists of departing from the same location with the same initial orientation but arriving at the same position with different orientations. Both simulations and tests on robot have been carried out.

In the simulation, robot is assigned to move toward the target starting from the same point [9.46975971, -45.203681, -7.69090778] with the same orientation [0.1294, -0.4551, -0.5036] and adapts its gripper orientation to 3 desired orientations: [0, -90, 0], [-90, 0, 90], [-135, 0, 90] within a displacement of more than 0.62m. These parameters will pass into 6D reach GFN based on motion loop algorithm in II-E, together with initial

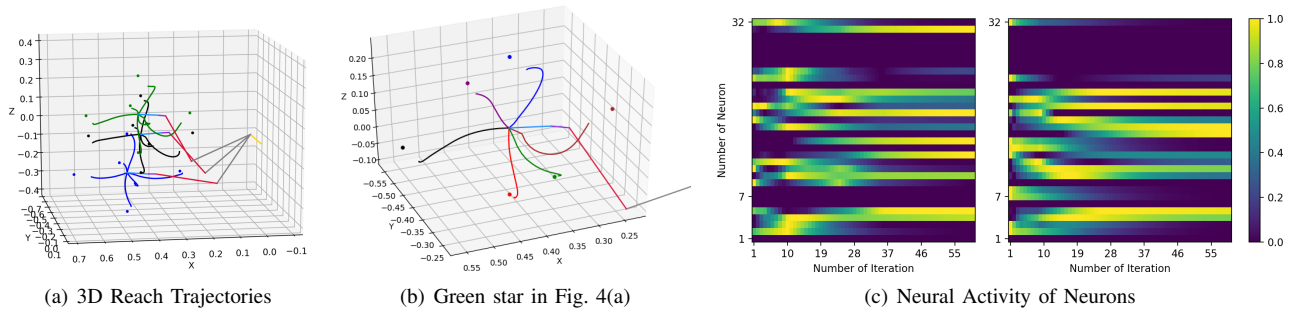


Fig. 4. 3D reaching task. (a) Robot arm starts from 3 different positions and approached 6 targets placed evenly around them at a distance of 0.2 meters. (b) Green star path in (a). (c) Neuron activities when going towards X+ direction (Left) and X- direction (right).

posture obtained directly from Reachy. Results are shown

TABLE III
6D REACH EXPERIMENTAL RESULTS

	[0, -90, 0]	[-90, 0, -90]	[-45, 0, 90]
Iteration	30	28	60
Final Distance Error	0.0143	0.0292	0.0432
Initial Orientation Error	46.43	146.56	120.64
Final Orientation Error	16.51	11.94	19.38

in Fig. 5 in terms of the robot motion path across iterations. Red arrows represent orientations of the gripper under three status: initial, final and desired, where the initial one and the final one are with corresponding annotations “Start” and “Target”. For a better representation, arrows in green, blue and black form a robot gripper coordinate frame. As can be noted, the 3D orientation adaptation of the gripper occurred since the beginning of 3D reach. Additionally, experimental results are detailed in Table. III, where orientation errors are in degree and distance errors are in meter.

From these results, we conclude that despite the displacement vectors being the same, when the desired orientation, as supplemental information mentioned previously, changes, GF neurons are sensitive to categorize this subtle adjustment and then interact with all arm joints to perform the movement, instead of tuning independently wrist displacements as ‘separate subsystems’ from ‘approach scheme’ and ‘rotate schema’ [12], which is deemed to be more human-like. Additionally, similar tests on real robot Reachy with desired orientations listed in the footnote of Table I have been recorded in video⁵ with figures of how populations of neurons discharged over time. Again, we can see how 6D reaching neurons modulate the robot arm to reach targets with the desired orientations under the real circumstance.

D. Learning Transformation of External Vision of a Tool

Two previous experiments show the effectiveness of GFNs in the 3D reach and 6D reach task. Differently, however, tool use implies advanced cognitive skills and abstraction abilities as the *Vision* is in the tool-centered reference frame, external to the body [31]. Therefore, in this experiment, we

investigate how Gain-field neurons react toward such tasks through simulations and tests on the real robot Reachy.

Similar to III-B, the task consisted of reaching 6 target points located in a 3D six-angled star with an internal angle of 90° that were equally 20 cm distant from the central starting point to determine if the robot was able to reach targets placed in all different directions. The nuance is that, in this case, the robot interacts with tools to perform this reach. Therefore, visual input becomes the displacement vector of tool tip instead of the one of end-effector. Likewise, the motor command generation is done using Eq. 4.

Three tool-use reaching tasks were carried out and results are represented in three different colors in Fig. 6(a), corresponding to three initial Cartesian coordinates: [0.455, -0.175, 0.025], [0.414, -0.343, 0.180], [0.398, -0.329, 0.283]. Additional individual view of each trajectory is shown in Fig. 6(b) - Fig. 6(d). The tool is drawn in orange and the 8 DoF robot arm is plotted in gray. Blue, red, green, purple, black and brown curves correspond to the trajectories for going towards Z+, Z-, Y+, Y-, X+ and X- respectively. In sum, the robot is able to start from the desired direction and interact with the stick to finally approach targets within 60 iterations with the precision around 0.017m on average.

Further demonstrations with real robot can be found online⁶. It is noticeable that tools are not always in parallel with each other, slight movements on tool tip are carried out during approach. Thus we claim that rather than keeping the tool always parallel, GF neurons have the capacity of devising specific transformations to organise a sequence of active human-like moves to reach targets.

To visualize performances across two reach networks, complementary trials that robot Reachy moves on the same path with and without the tool have been done. Similarly, initial postures, starting and end position are fed to two networks. Predictions returned by networks are then executed by Reachy. Video of this experiment can be found on our site.⁷ As one can see, by starting from the same position,

⁵[https://perso-etis.ensea.fr/neurocyber/Videos/authors/xiaodan/Grasp Experiment.mp4](https://perso-etis.ensea.fr/neurocyber/Videos/authors/xiaodan/Grasp%20Experiment.mp4)

⁶<https://perso-etis.ensea.fr/neurocyber/Videos/authors/xiaodan/Simulation.tool-use.mp4>

⁷[https://perso-etis.ensea.fr/neurocyber/Videos/authors/xiaodan/Tool-use Experiment.mp4](https://perso-etis.ensea.fr/neurocyber/Videos/authors/xiaodan/Tool-use%20Experiment.mp4)

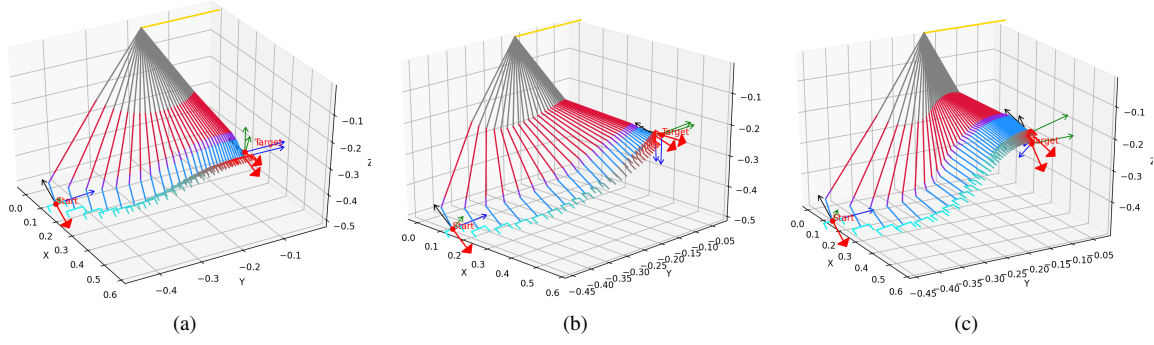


Fig. 5. 6D reach task with orientation: (a) $[0^\circ, -90^\circ, 0^\circ]$ (b) $[-45^\circ, 0^\circ, 90^\circ]$ (c) $[-45^\circ, 0^\circ, 90^\circ]$. Red arrows represent orientations of the gripper under three different statuses: initial, final and desired, where the initial one and the final one are with corresponding annotations “Start” or “Target”. Arrows in green, blue and black form a robot gripper coordinate frame.

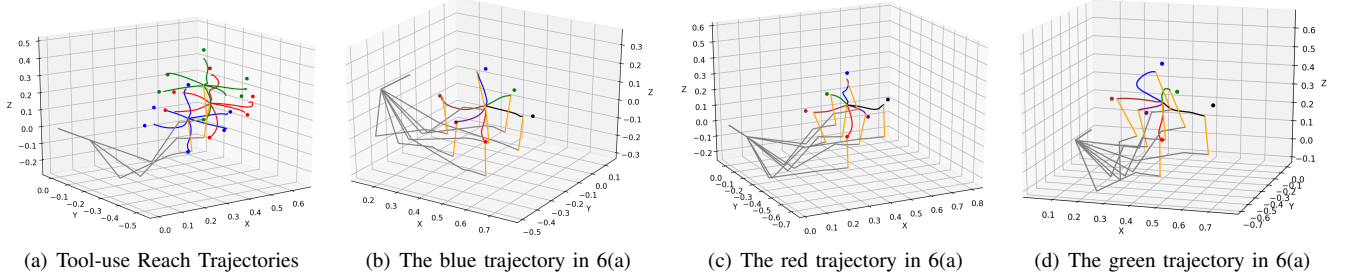


Fig. 6. The robot arm and the tool are drawn in gray and orange respectively.

the robot hand and the tool gripped by it can arrive at the defined position based on the estimation return by the previous reach and tool-use networks. By implementing these two tests together, we aim to elucidate that firstly, GF neurons succeed in mapping external information on internal information through multisensory processing. Secondly, such a multiplication permits a rapid learning of this altered relationship between the stick movement and the hand movement (forward/inverse kinematics models of the new utensil) by minimizing continuously errors between them, thus facilitating the learning of new tasks [20].

IV. CONCLUSION

In this paper, we suggested biological neural models for efficient learning of 3D reach, 3D reaching + 3D orientation and tool-use reach in 3D space through the remapping of multi-modalities based on the mechanism of gain-modulation, which is in line with the literature in cognitive neuroscience. Experimental results in III reveal that GFNs permit efficient learning of 3D reach, 6D reach and tool-use reach in 3D space. Precisely, III-A suggests that GF neurons are sensitive to visual directions when reaching. Such a preference on the direction during movement help elucidate the underlying mechanism on generating motor schema, as mediating excitation and inhibition of GF reaching/grasping neurons with respect to *proprioceptive* and *visual* information. The 3D reaching neural model in section III-B was developed to evaluate the GFN’s performance in remapping the *vision* (e.g., movement vector) as well as *proprioception* (e.g., initial posture) to *motion* (e.g., final

posture). Its promising results constitute the basis for the next two more complex manipulations. Therefore, in III-C, orientation as supplemental information is integrated with displacement direction into *vision*. In our experiment, gain-field 6D reaching neurons are able to give rise to a “subtle adjustment” thus actuate continuously and constantly all arm joints to approach with prescribed orientations, without being divided into “transport” and “preshape” processes, which is in accordance with [35] and the long movement prediction model in [38]. Additionally, in III-D, we extend the GFN to conduct simple tool-use actions where *vision* is no longer in the egocentric reference frame. The results show that GF neurons have competency on such a nonlinear mapping of external visual information to self-body dynamics to construct internal models of new tools of the external world [30], [20].

With these properties of GFNs, we believe that the GF neurons are capable of aligning the hand and wrist to objects and thus effective enough for permitting 3D, 6D and tool-use reach and even new tasks with higher degrees of freedom robotic system without relearning from scratch. This approach can serve for the continual learning of increasingly complex tasks binding of separated features in the sensorimotor networks. In the future, we will exploit inter-personal interactions such as imitation between cobots, learning of a coordinated opening of the gripper and more complex manipulation of multi fingers ambidextrous robot gripper with gain-field networks.

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